

Generalized method of a resolvent operator expansion. III

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We show that our Born-like parametrized expansion of $R(E) = (E - H)^{-1}$ is, after minor modifications, well defined even at the pole of $R(E)$ and provides a new method of solving linear homogeneous equations. Three different possibilities of application are discussed here:

- (1) An analytic method of solving the differential equations. It is based on the partitioning of generalized power series and illustrated by the new solution of the s -wave Schrödinger equation.
- (2) A consequent model space reduction of Schrödinger equation. In terms of the matrix moments of H , the effective interaction is defined as an operator continued fraction.
- (3) A new form of the perturbation theory which dispenses with the solution of the unperturbed problem.

1. INTRODUCTION

There exists a particularly compact unification of various perturbation theories based on the search for the poles $E = \lambda_0$ of the resolvent operator (Green function) $R(E) = (E - H)^{-1}$. It is described, e.g., by Kumar¹ and consists of the formal factorization of $R(E)$ into the singular and regular parts $W(E)$ and $\mathcal{M}(E)$, respectively, in the vicinity of the pole. Putting

$$W(E) = \text{const}/\det(E - H), \quad (1.1)$$

we may also include some numerical diagonalization methods² in this scheme since the projector which defines an eigenstate of full ("perturbed") Hamiltonian H coincides with the regular part $\mathcal{M}(E)$ of $R(E)$ in the limit $E \rightarrow \lambda_0$. In the present paper, we also include in the same scheme our method of expansion of $R(E)$ (Refs. 3 and 4 quoted here as I and II, respectively). In this way we obtain a new, very flexible method of solving linear homogeneous equations

$$(E - H)|\psi\rangle = 0. \quad (1.2)$$

The corresponding modified form of expansion is described in Sec. 2. In the spirit of the method I, we interpret Eq. (1.1) simply as an additional restriction for the choice of the free parameters. The necessity of such a restriction follows from the fact that after an arbitrary (parametrized) rearrangement of the Born series (I), we cannot arrive at the convergent representation of $R(E)$ near the pole. As a consequence of the restriction, the natural factorization [definition of $W(E)$] immediately follows and uniquely defines $\mathcal{M}$.

An example showing how the method works in its generality is given in Sec. 3. The new analytic solution obtained for the s -wave Schrödinger equation with p exponentials, $p > 1$, has a form of a generalized Bessel function.

When the results of II are taken into account, the projector operator formalism leads to the more restrictive specification of parameters. In this spirit, we complete the standard model space reduction of the Schrödinger equation by the operator continued fraction definition of the effective interaction. The formalism assumes the knowledge of the model space moments $\langle m \rangle = PH^mP$ of H and is described in Sec. 4.

There is an interesting application of the moment meth-

od of Sec. 4 based on the Taylor series expansion of the moments $\langle m \rangle$. This generalizes the standard perturbation techniques and is developed in Sec. 5.

Finally, in Sec. 6 we discuss the numerical aspects and possible applications of the methods as algorithms for the diagonalization of some special matrices.

2. BORN-LIKE EXPANSION FOR BOUND STATES

Let us write the Hamiltonian H in the representation

$$H = \sum_{\lambda} |\lambda\rangle \lambda \langle\lambda| + H_c, \quad (2.1)$$

where the $|\lambda\rangle$'s correspond to bound states ($\lambda < 0$) and H is the continuous spectrum part of H . In the vicinity of $\lambda = \lambda_0$, the pole term in the resolvent operator $R(E) = (E - H)^{-1}$ dominates so that the regularized operator

$$(E - \lambda_0)R(E) = \sum_{i=1}^{m(\lambda_0)} |\lambda_{0i}\rangle \langle\lambda_{0i}| + O(E - \lambda_0) \quad (2.2)$$

approximates the projector on bound states $|\lambda_{0i}\rangle$ corresponding to the energy λ_0 . We intend to apply in Eq. (2.2) the rearranged Born expansion I.

We must extend its validity to the singular point λ_0 . Introducing the functions (vectors) $|Y_0^i(E, \lambda_0)\rangle$ by the equation

$$(E - H)|Y_0^i(E, \lambda_0)\rangle = (E - \lambda_0)|X_1^i\rangle, \quad i \in 1, M_1, \quad (2.3)$$

we may write formally (for $E \neq \lambda_0$, cf. I)

$$\begin{aligned} |Y_0^i(E, \lambda_0)\rangle &= (E - \lambda_0)R(E)|X_1^i\rangle \\ &= (E - \lambda_0) \sum_{l=1}^{\infty} \sum_{s=1}^{M_l} D_{il}^{is}(E) |X_s^l\rangle, \end{aligned} \quad (2.4)$$

$$D_{il}^{is}(E) = \sum_{\substack{s_j = 1, i_{j+1} = 1 \\ j = 1, 2, \dots, l-1}}^{M_l} \sum_{j=1}^{M_{l-1}} F_1^{is_1}(E) B_1^{s_1 i_2} F_2^{i_2 s_2}(E) \dots B_{l-1}^{s_{l-1} i_l} F_l^{i_l s}(E).$$

Here, the $(M_k \times M_k)$ -dimensional matrix continued fractions (MCF) $F_k(E)$ are defined by the recurrence

$$F_k(E) = [EI - A_k - B_k F_{k+1}(E) C_{k+1}]^{-1}, \quad k = 1, 2, \dots, \quad (2.5)$$

with $(M_k \times M_k)$ -, $(M_k \times M_{k+1})$ -, and $(M_{k+1} \times M_k)$ -dimen-

sional matrices of free parameters A_k , B_k , and C_{k+1} , respectively, $M_k < \infty$. The relation

$$H|X_k^s\rangle = \sum_{r=1}^{M_k-1} C_k^{sr}|X_{k-1}^r\rangle + \sum_{r=1}^{M_k} A_k^{sr}|X_k^r\rangle + \sum_{r=1}^{M_{k+1}} B_k^{sr}|X_{k+1}^r\rangle, \quad k=1,2,\dots, \quad (2.6)$$

specifies (possibly nonuniquely) the expansion set $\{|X_k^r\rangle\}$ in terms of H and free parameters.

Now, we consider the limiting transition $E \rightarrow \lambda_0$. Because the vector

$$|Y_0^i(E, \lambda_0)\rangle \rightarrow \sum_{\substack{j=1 \\ \lambda_0 = \lambda_{0j}}}^{m(\lambda_0)} |\lambda_{0j}\rangle c_j^i, \quad c_j^i = \langle \lambda_{0j} | X_1^i \rangle \quad (2.7)$$

becomes equal to the solution of the homogeneous equation

$$(E - H)|Y_0^i(E, \lambda_0)\rangle = 0 \quad (2.8)$$

for $E = \lambda_0$, our aim is to give the precise meaning to the $E \rightarrow \lambda_0$ limit on the right-hand side of Eq. (2.4). It is clear that the difficulties are similar to those in the Born series⁵: The existence of the auxiliary sequence F_k , $k=1,2,\dots,N$, for any cutoff $N < \infty$ implies that the expansion (2.4) ceases to converge because $(E - \lambda_0)D_{1l}(E) \rightarrow 0$ and $|Y_0^i(E, \lambda_0)\rangle - (E - \lambda_0)\Sigma_{l,s} D_{1l}^{is}|X_1^s\rangle \rightarrow |Y_0^i(\lambda_0, \lambda_0)\rangle$ for $E \rightarrow \lambda_0$.

We shall show first that it is sufficient to demand the existence of F_k 's for $k \geq 2$ only. Really, putting

$$|Y_0^i(E, \lambda_0)\rangle = \sum_{s=1}^{M_1} \mathcal{N}_0^{is} |\psi_0^s(E)\rangle, \quad |\psi_0^i(E)\rangle = \sum_{l=1}^N \sum_{s=1}^{M_l} d_l^{is}(E) |X_1^s\rangle, \quad (2.9)$$

$$\mathcal{N}_0^{is} = (E - \lambda_0)F_1^{is}(E),$$

$$d_l(E) = B_1 F_2(E) \dots B_{l-1} F_l(E),$$

$$l=2,\dots,N=\infty, \quad E \neq \lambda_0, \quad d_1(E) = I, \quad i \in 1, M_1,$$

and assuming

$$0 = \det[\lambda_0 I - A_1 - B_1 F_2(\lambda_0) C_2] = \det\{[F_1(\lambda_0)]^{-1}\}, \quad (2.10)$$

we see that the limit $(E - \lambda_0)F_1(E) = 0 \times \infty$ is indeterminate and hence $|Y_0^i(\lambda_0, \lambda_0)\rangle$ may become nonzero.

The simpler regularization of the series $|Y_0^i(\lambda_0, \lambda_0)\rangle$ may be achieved in analogy with the Lanczos numerical method⁶: We replace $\mathcal{N}_0$ in Eq. (2.9) [and Eq. (2.4)] by the energy-independent normalization matrix $\mathcal{N}$ that satisfies the condition

$$\mathcal{N} \cdot [\lambda_0 I - A_1 - B_1 F_2(\lambda_0) C_2] = 0. \quad (2.11)$$

It follows from Eq. (2.10) that $\mathcal{N} \neq 0$ exists. In fact, the substitution $\mathcal{N}_0 \rightarrow \mathcal{N}$ corresponds just to another (energy-dependent) normalization chosen on the right-hand side of Eq. (2.3) from the very beginning.

Of course, the new form

$$(E - H)|Y^i(E, \lambda_0)\rangle = \sum_{r,s=1}^{M_1} \mathcal{N}^{ir} [F_1^{-1}(E)]^{rs} |X_1^s\rangle \quad (2.12)$$

of Eq. (2.3) represents the same homogeneous Eq. (2.8) or (1.2) in the limit $E \rightarrow \lambda_0$.

The multiplicity $m'(\lambda_0)$ of the root λ_0 in Eq. (2.10) determines the possible number of independent rows in the matrix $\mathcal{N}$. We may therefore conclude the following:

Proposition: Provided that

- (D1) the auxiliary sequence of inverted matrices $[F_k(E)]^{-1}$ is uniquely defined for all $k \geq 1$ by the free parameters according to Eq. (2.5),
- (D2) the relation (2.6) between H and $|X\rangle$'s holds,
- (D3) the series $|\psi_0^i(\lambda_0)\rangle$ defined in Eq. (2.9) converges, and

(D4) the multiplicity $m'(\lambda_0) \leq M_1$ of the root in Eq. (2.10) is less than or equal to the multiplicity $m(\lambda_0)$ of the same eigenvalue λ_0 of H , then, with $m'(\lambda_0)$ independent rows of the "normalization" matrix $\mathcal{N}^{is}$, $i \in 1, m'(\lambda_0)$,

$s \in 1, M_1$, subjected to the finite number of linear algebraic Eqs. (2.11), the vectors

$$|Y^i(\lambda_0, \lambda_0)\rangle = \sum_{j=1}^{M_1} \mathcal{N}^{ij} |\psi_0^j(\lambda_0)\rangle, \quad i \in 1, m'(\lambda_0) \quad (2.13)$$

represent the $m'(\lambda_0)$ independent solutions of the homogeneous Eq. (1.2) corresponding to the eigenvalue λ_0 .

Remarks:

Ad (D1): When the convergence of MCF (2.5) cannot be proved for our choice of free parameters, we may still employ an alternative (polynomial) auxiliary sequence Γ (with possibly different convergence domain) introduced in Appendix II of Ref. 7.

Ad (D2): The main assumption defining our method is just the "clever" fulfillment of the relation (2.6) which represents the generalization of the Lanczos recurrent construction of the basis (cf. Refs. 2, I, II and Secs. 3 and 6).

Ad (D3): Again, the typical proof of convergence is given in Sec. 3. Another (iterative) approach was described in detail in Ref. 7.

Ad (D4): This is the only assumption pertinent to the $E = \lambda_0$ case of our expansion method. It restricts further the freedom in the choice of free parameters (e.g., fixing ξ in Sec. 3). In this respect, it is not independent of (D3) (cf. how the knowledge of some eigenstates of H is used for the acceleration of convergence of the Born series⁵) and influences also (D2) and (D1) (both F_k and $|X\rangle$ depend on the parameters).

3. APPLICATION TO DIFFERENTIAL EQUATIONS: NEW ANALYTIC SOLUTION OF THE SCHRÖDINGER EQUATION

Certain classes of differential equations may be solved by the Taylor series ansatz leading to the solvable two-term recurrence relations for coefficients.⁸ The matrix generalization of these relations enables us to cover the more complicated differential equations by the analogous ansatz which may contain more free parameters corresponding to the constants in the differential equation. We present an illustrating example in what follows.

In the Schrödinger equation (1.2) we consider the Hamiltonian acting in the s wave by the superposition of the exponential potentials,

$$H = H(p) = -\frac{d^2}{dr^2} - \sum_{i=1}^p g_i e^{-\gamma_i r}, \quad (3.1)$$

where $0 < \gamma_1 < \gamma_2 < \dots < \gamma_p$, $0 < r, g_i \in \mathbb{C}$. This form of the potential [discrete Laplace transform of an arbitrary function $g(\gamma)$] may be used in many realistic situations. The solution with $p = 1$ is known. It is represented by the Bessel function.⁵ For $p > 1$, the solution was known only in an implicit form.^{5,9}

We apply the method of Sec. 2. The relation (2.6) may be fulfilled when we use $|X_k^m\rangle$ in the form of exponentials

$$\langle r|X_k^m\rangle = e^{-\varphi(k,m,\xi)r}. \quad (3.2)$$

For simplicity we shall number them by the p nonnegative integers $i_1, i_2, \dots, i_p$ and define

$$\begin{aligned} \varphi(k,m,\xi) &= \xi + \sum_{n=1}^p i_n \gamma_n, \\ k &= 1 + \sum_{n=1}^p i_n \leq N = \infty, \end{aligned} \quad (3.3)$$

$$\begin{aligned} m &= 1 + \binom{i_1}{1} + \binom{i_1 + i_2 + 1}{2} + \dots \\ &\quad + \binom{k-1-i_p+p-2}{p-1} \\ &\leq M_k = M_k(p) = \binom{k+p-2}{p-1}, \end{aligned}$$

where ξ is a free (complex) parameter. The one-to-one mapping $\{i_n\}_{n=1}^p \leftrightarrow (k,m)$ may be inverted by the prescription

$$i_p = \max j, \quad \binom{k+p-3-j}{p-1} \leq m-1,$$

with the $(p-2)$ times repeated formal substitution

$$\begin{aligned} m_{\text{new}} &= m - \binom{k+p-3-i_p}{p-1} \geq 1, \\ k_{\text{new}} &= k - i_p \geq 1, \quad p_{\text{new}} = p - 1 \geq 1. \end{aligned}$$

Equation (2.6) determines the form of the parametric matrices

$$\begin{aligned} C_{k+1} &= 0, \quad A_k = A_k^{ml}(p) = -\delta_{ml} \varphi^2(k,m,\xi), \\ F_k(E) &= F_k^{ml}(E,p) = \delta_{ml} [E + \varphi^2(k,m,\xi)], \\ m, l &\in \overline{1, M_k(p)}, \end{aligned} \quad (3.4)$$

$$\begin{aligned} B_k &= B_k(p) \\ &= \begin{pmatrix} -g_p I_1 & B_1(p-1) & 0 & \dots & 0 \\ 0 & -g_p I_2 & B_2(p-1) & & 0 \\ \dots & & & & \\ 0 & \dots & 0 & -g_p I_k & B_k(p-1) \end{pmatrix} \end{aligned}$$

$p > 1$,

$$\begin{aligned} B_k(1) &= -g_1, \quad I_k = I_k^{ml}(p) = \delta_{ml}, \quad m, l \in \overline{1, M_k(p-1)}, \\ k &= 1, 2, \dots, \end{aligned}$$

so that the conditions (D1) and (D2) of the Proposition are satisfied. Since $M_1(p) = 1$, it is easy to satisfy (D4) when we specify $E = \xi^2$. To verify the Proposition and obtain a formal solution of the Schrödinger equation, it remains to prove the convergence (D3) of the series $|\psi_0\rangle$ for $N = \infty$.

Lemma: The series

$$\begin{aligned} \langle r|Y\rangle &= \mathcal{N} \sum_{l=1}^N \sum_{i=1}^{M_l} d_l^i(\xi,p) e^{-\varphi(l,i,\xi)r}, \\ N &= \infty, \quad \mathcal{N} = d_1^1(\xi,p) = 1, \end{aligned} \quad (3.5)$$

$$d_l^i(\xi,p) = \sum_{i=1}^{M_{l-1}} d_{l-1}^i(\xi,p) B_{l-1}^{ij}(p) F_l^{ij}(-\xi^2,p),$$

$$j \in \overline{1, M_l}, \quad l \in \overline{2, N},$$

i.e., Eq. (2.9) written in terms of functions (3.2) and parameters (3.4) is defined and convergent for arbitrary complex $\xi = (-E)^{1/2} \neq \xi(k,m) = -\sum_{i=1}^p i_i \gamma_i / 2$, $k > 1$, and bounded for $\text{Re} \xi > 0$.

Proof: We prove the absolute convergence of $\langle r|Y\rangle$ by majorizing the series (3.5) and avoiding the points $\xi(k,m)$ where $\det F_k^{-1} = 0$, $k > 1$, and the matrix F_k ceases to exist. For $r \in (r, \infty)$, we have $|\langle r|X_k^m\rangle| \leq \exp(-r \text{Re} \xi)$ so that we may put $r = 0$ without loss of generality. For large k , we replace the elements of matrix F_k by the expression $(k\gamma_1)^{-2} + O(k^{-3})$. Thus the action of $B_{k-1} F_k$ on $\langle 0|X_k^m\rangle$ may be represented by identity matrix multiplied by the overall numerical factor $h k^{-2} + O(k^{-3})$, $h = \sum_{n=1}^p (-g_n) / \gamma_1^2$ since the number of nonzero elements in each row of B_k is equal to p . The majorizing series may therefore be constructed when we substitute $|g_i|$ for $-g_i$ in B_k and h . This is the power series in h with an infinite radius of convergence. QED

Consequence: Provided that the choice of the free parameter $\xi > 0$ is given by the transcendental equation

$$0 = 1 + \sum_{l=2}^N \sum_{i=1}^{M_l} d_l^i(\xi,p), \quad N = \infty, \quad (3.6)$$

and the normalization coefficient (number) $\mathcal{N}$ is defined by the expression

$$\begin{aligned} \mathcal{N}^{-2} &= \sum_{k,k'=1}^N \sum_{l=1}^{M_k} \sum_{l'=1}^{M_{k'}} d_k^l(\xi,p) d_{k'}^{l'}(\xi,p) \\ &\quad \times [\varphi(k,n,\xi) + \varphi(k',n',\xi)]^{-3}, \end{aligned} \quad (3.7)$$

the series $\langle r|Y\rangle$ represents a bound state corresponding to the s -wave Hamiltonian (3.1) and energy $E = -\xi^2 < 0$.

Proof: Equation (3.6) represents the standard boundary condition $\langle r|Y\rangle = 0$ for $r = 0$ and $\xi > 0$ guarantees the integrability for $r = \infty$. From the definition $\langle Y|Y\rangle = 1$ with the insertion of formula¹⁰ $\langle X_k^m|X_{k'}^{m'}\rangle = [\varphi + \varphi']^{-3}$ we get Eq. (3.7). QED

In the known special case $p = 1$, the solution $\langle r|Y\rangle$ with the normalization Eq. (3.7) coincides with the Taylor series for the Bessel function^{5,10} $J_{2\xi/\gamma}(2g^{1/2} \exp(-\gamma r/2)/\gamma)$. For $p > 1$, the analogous compact form of solution $\langle r|Y\rangle$ seems to be new (cf. Refs. 5, 9).

Equation (3.6) defines the binding energies as zeros of transcendental functions. We have verified numerically the consistency and stability of such a procedure. Putting $\gamma_i = \gamma$ and $g_i = g/p$, $i \in \overline{1,p}$ for $p = 1, p = 2$, and $p = 4$, we have reproduced, within the range of the rounding errors, the known eigenvalues of the classical exponential potential. The convergence of partial sums ($N < \infty$ approximations) was also good for different samples of potentials.

With $\mathcal{N} = 1$, $\xi = -iE^{1/2}$, $E > 0$, the function $\langle r|Y\rangle$

represents the irregular solution of Eq. (1.2) and $\langle 0|Y\rangle$ is the Jost function.⁵ Thus, the use of the pure imaginary values of ξ provides the scattering solutions and may be useful in the realistic fit of any s -wave phase shifts by the exponential potentials (couplings g_i and ranges γ_i , $i \in 1, p$).

4. MCF DEFINITION OF THE EFFECTIVE INTERACTION IN THE SCHRÖDINGER EQUATION

We assume that the exact Schrödinger equation (1.2) may in principle be investigated by means of the limiting transition $N \rightarrow \infty$ in its separable approximation

$$I_N H I_N |\psi_N\rangle = E_N |\psi_N\rangle, \quad (4.1)$$

where N denotes the cutoff in the approximate identity projector

$$I_N = \sum_{k=1}^N P_k, \quad P_k = \sum_{i,j=1}^M |X_k^i\rangle (D_k^{-1})^{ij} \langle X_k^j|, \quad (4.2)$$

$$\langle X_k^i | X_k^j \rangle = \delta_{ik} \cdot D_k^{ij},$$

and $I_\infty = 1$. Further, we assume that this transition $N \rightarrow \infty$ is technically impossible (e.g., due to the computer limitations) and that only the M -dimensional ($M \ll 0$) model space $\mathcal{H}_P$ specified by the projector $P = P_1$ is tractable in practice. It is known¹¹ that the exact solution of linear Eq. (4.1) is still possible within the model space $\mathcal{H}_P$ because the nonlinear equation

$$\begin{aligned} [E_N - PHP - V_N(E_N)]P|\psi_N\rangle &= 0, \\ V_N(E) &= PHQ_N \frac{1}{E - Q_N H Q_N} Q_N H P, \\ Q_N &= I_N - P = P_2 + P_3 + \dots + P_N, \end{aligned} \quad (4.3)$$

accompanied by the definition

$$Q_N |\psi_N\rangle = (E_N - Q_N H Q_N)^{-1} Q_N H P |\psi_N\rangle,$$

is fully equivalent to Eq. (4.1). Only the definition of the so-called effective Hamiltonian $H_{\text{eff}} = PHP + V_N(E)$ necessitates leaving the model space.

In the spirit of Sec. 2, Eq. (4.1) may be solved by the MCF method. Using the (generalized) Lanczos idea of generating $|X_k\rangle$'s by H , and their elimination for $k = N, N-1, \dots, 2$ according to II, we are able to specify the resulting equation [Eq. (2.11)] in terms of the model space quantities only. It is exciting that the change of notation "translates" Eq. (2.11) just into Eq. (4.3) where the effective interaction correction $V_N(E)$ is redefined and given as the model space $[(M \times M)$ -dimensional, $M \ll \infty$], recursively defined operator. In this way the manageable explicit MCF formulas for the effective interaction are obtained not necessitating to leave the model space.

This idea is not entirely new because it coincides with the classical moment method¹² for $M = 1$. The great advantage of using $M \gg 1$ lies in the sufficiency of low N 's to determine the corrections to Eq. (4.1). Even the lowest $N = 2$ case of Eq. (4.1) with the effective interaction correction

$$V_2(E) = \beta_2 \frac{1}{E\beta_2 - \alpha_2} \beta_2,$$

$$\beta_2 = PH^2P - PHP \cdot PHP = PHQHP, \quad Q = 1 - P = Q_\infty, \quad (4.4)$$

$$\begin{aligned} \alpha_2 &= PH^3P - PH^2PPHP - PHPPH^2P + PHPPHPHP \\ &= PHQHQP \end{aligned}$$

provides a useful alternative to the standard doubling of dimension of arbitrary basis in Eq. (4.1). Provided that the correction $\epsilon_{21} = E_2 - E_1$ is small, it is also easy to show that the first-order Taylor expansion of the secular determinant leads to the compact approximate prescription

$$\begin{aligned} E_2 &= E_1 - 1/\text{tr}[G(E_1)] + O(\epsilon_{21}^2), \\ G(E) &= [1 + V_2(E)(1/PHQHP)V_2(E)] \\ &\quad \times [E - PHP - V_2(E)]^{-1} \end{aligned} \quad (4.5)$$

for an improved energy.

An entirely analogous procedure is to be applied for any higher $N > 2$. The algorithm reads:

(a) Calculate the model space moments $\langle m \rangle = PH^m P$, $m \in 1, 2N-1$ of the Hamiltonian operator H .

(b) Evaluate the auxiliary matrices α_k, β_k from the formulas in II using the auxiliary sequences $\eta_k^{[l,r]}, \theta_k^{[l,r]}$.

(c) Find the solution of Eq. (4.3) where the explicit MCF formula for the effective interaction correction (cf. II) reads

$$\begin{aligned} V_N(E) &= \beta_2 w_2(E) \beta_2, \\ w_k(E) &= [E\beta_k - \alpha_k - \beta_{k+1} w_{k+1}(E) \beta_{k+1}]^{-1}, \\ &\quad k \in 2, N. \end{aligned} \quad (4.6)$$

Besides the trivial initialization we may also use an improved one,¹³ $w_{N+1} \neq 0$, based on our physical interpretation of P_{N+1} .¹⁴

It is useful to use some symbolic manipulation language and the computer-produced codes since the explicit formulas become quite involved even at the next $N = 3$ cutoff for $D = I$ where, $g_2 = 1/\beta_2$, $\langle m \rangle = \langle X_1 | H^m | X_1 \rangle$, and

$$\begin{aligned} \beta_3 &= \langle 4 \rangle - \langle 2 \rangle \langle 2 \rangle - (\langle 3 \rangle - \langle 2 \rangle \langle 1 \rangle) g_2 (\langle 3 \rangle - \langle 1 \rangle \langle 2 \rangle), \\ \alpha_3 &= \langle 5 \rangle - \langle 2 \rangle \langle 1 \rangle \langle 2 \rangle - (\langle 3 \rangle - \langle 2 \rangle \langle 1 \rangle) g_2 \\ &\quad \times (\langle 4 \rangle - \langle 1 \rangle \langle 1 \rangle \langle 2 \rangle) - (\langle 4 \rangle - \langle 2 \rangle \langle 1 \rangle \langle 1 \rangle) g_2 \\ &\quad \times (\langle 3 \rangle - \langle 1 \rangle \langle 2 \rangle) + (\langle 3 \rangle - \langle 2 \rangle \langle 1 \rangle) g_2 \\ &\quad \times (\langle 3 \rangle - \langle 1 \rangle \langle 1 \rangle \langle 1 \rangle) g_2 (\langle 3 \rangle - \langle 1 \rangle \langle 2 \rangle). \end{aligned}$$

By this complexity (and by the computer time consumption) we pay for lowered dimension of P with respect to I_N (saving computer storage). The comparable complexity of formulas survives even at $M = 1$, where the diagrammatic methods are used for α_k, β_k and their evaluation,¹⁵ and it increases with N . We may say that without the significant lowering of the cutoff N (and hence the number of moments) by increasing M (dimension $M \times N$ of I_N is fixed), the method would suffer also from cancellations (loss of precision in the computer), and in the classical $M = 1$ case it seems to be quite impracticable [cf. paragraph (g) in Sec. 6].

While the energy E_N is a good approximation to the exact $E = E_\infty$ for sufficiently high M, N , only the M -dimensional projection $P|\psi_N\rangle$ of the full solution $|\psi_N\rangle$ is obtained from Eq. (4.3) for any N . If $Q|\psi_N\rangle$ cannot be neglected, we should not eliminate the rest of the basis. More in the spirit of the moment approach is the repartitioning of the basis such

that $|X_i^j\rangle_{\text{new}}, i \in 1, M_{\text{new}}$, will include further interesting components from $|X_k^j\rangle_{\text{old}}, k > 1$. Because an increase of dimension M would make the evaluation of the matrix continued fraction $w_2(E)$ too time-consuming, the dimension reduction is to be used. It is based on the investigation of β_k 's and resembles (generalizes) one of the practical advantages of the standard Lanczos technique, namely the natural truncation of the basis.^{2,16} Really, once $\beta_k = \langle \xi_k | \xi_k \rangle$ (cf. II) is singular ($\det \beta_k = 0$), it is obvious that the set of $M = M_{\text{old}}$ Lanczos vectors $|\xi_k\rangle (= |X_k\rangle$ without normalization) is linearly dependent. The unitary $M \times M$ transformation U of $|\xi_k\rangle$ is to be found (e.g., by diagonalization of β_k) which annihilates the last vector, $|\xi_k^M\rangle_{\text{new}} \equiv 0$. The same U transformation applied to all vectors simultaneously, enables us either to cross out $|X_1^M\rangle_{\text{new}}$ and start, from the very beginning, with $M_{\text{new}} = M - 1$ initial vectors $|X_i^j\rangle_{\text{new}} = \sum_{j=1}^M |X_i^j\rangle_{\text{old}} U^{ji}$, $i \in 1, M_{\text{new}}$, or rather lower just the dimension of higher partitions, $M_{k+i} = M - 1, i \geq 0$, simplifying the structure of MCF.

In principle, the elimination of the (hidden) dependence of the initial vectors $|X_i^j\rangle$ may occur repeatedly. In full analogy with the Lanczos $M = 1$ case, we may arrive at the finitization of the basis (i.e., $\beta_N \equiv 0$ at some N) provided that all the initial vectors $|X_i^j\rangle, i \in 1, M_1, M_1 < \infty$, span only a finite invariant space of H . In practice, even the smallness of $\det \beta_k$ should fully be employed in lowering the dimension M thus providing the most natural truncation procedure at all.

5. GENERALIZED PERTURBATION THEORY

Knowledge of the moments $\langle m \rangle^{ij} = \langle X_i^j | H^m | X_i^j \rangle$, $i, j \in 1, M$ is sufficient for the method of the preceding paragraph to work. Since the calculation of moments $\langle m \rangle$ may represent the serious difficulty in practice, we shall now require that just the perturbation expansion ($\gamma \ll 1$)

$$\langle m \rangle = \langle m \rangle_{[0]} + \gamma \langle m \rangle_{[1]} + \gamma^2 \langle m \rangle_{[2]} + \dots \quad (5.1)$$

of moments is given. Let us investigate the consequences.

We apply the method of Sec. 4 to exact H denoting the perturbation order in all the formal expansions like Eq. (5.1) by adding the subscript in brackets. According to the general methodology described in II, we have first to construct the auxiliary sequence

$$\eta_k^{[l,r]} = \sum_{n=0}^{\infty} \gamma^n \eta_k^{[l,r,n]}, \quad l, r \in 1, k-1, \quad k = 1, 2, \dots$$

We put $\eta_k^{[l,r]} = \langle l+r \rangle_{[n]}, n = 0, 1, \dots$, and define

$$\begin{aligned} \beta_{k[n]} &= \eta_{k[n]}^{[k-1, k-1]}, \quad n = 0, 1, \dots, \\ g_{k[0]} &= 1/\beta_{k[0]}, \end{aligned} \quad (5.2)$$

$$g_{k[n]} = -g_{k[0]} [\beta_{k[1]} g_{k[n-1]} + \dots + \beta_{k[n]} g_{k[0]}], \quad n = 1, 2, \dots,$$

$$\eta_{k+1[n]}^{[l,r]} = \eta_{k[n]}^{[l,r]} - \sum_{n_1+n_2+n_3=n} \eta_{k[n_1]}^{[l, k-1]} g_{k[n_2]} \eta_{k[n_3]}^{[k-1, r]}, \quad n = 0, 1, \dots, \quad k = 1, 2, \dots,$$

in a recurrent way, so that the parameters $\beta_k = \sum_{h=0}^{\infty} \gamma^h \beta_{k[h]}$ are generated from Eq. (5.1). In an entirely

similar way we define also $\theta_{k[n]}^{[l,r]}$ and $\alpha_k = \sum_{n=0}^{\infty} \gamma^n \alpha_{k[n]}$ (cf. II). Then, the perturbation expansion of MCF in Eq. (4.6) leads to the general formula

$$\begin{aligned} w_k(E)_{[0]} &= [E\beta_{k[0]} - \alpha_{k[0]} - \beta_{k+1[0]} \\ &\quad \times w_{k+1}(E)_{[0]} \beta_{k+1[0]}]^{-1}, \\ w_k(E)_{[n]} &= -w_k(E)_{[0]} [\rho_{k[1]} w_k(E)_{[n-1]} \\ &\quad + \dots + \rho_{k[n]} w_k(E)_{[0]}], \\ \rho_{k[n]} &= E\beta_{k[n]} - \alpha_{k[n]} \\ &\quad - \sum_{n_1+n_2+n_3=n} \beta_{k+1[n_1]} w_{k+1}(E)_{[n_2]} \beta_{k+1[n_3]}, \\ n &= 1, 2, \dots, \quad k = N, N-1, \dots, 2, \end{aligned} \quad (5.3)$$

which defines simultaneously all the perturbation contributions in $w_k(E) = \sum_{n=0}^{\infty} \gamma^n w_k(E)_{[n]}$ by the polynomial recurrence. At the finite cutoff N , the trivial initialization $w_{N+1}(E)_{[n]} = 0, n = 0, 1, \dots$, defines, in a recurrent way again, the sequence $w_N(E), w_{N-1}(E), \dots, w_2(E)$. Finally, the perturbation expansion of the effective interaction correction

$$V_N(E) = \sum_{n=0}^{\infty} \gamma^n V_N(E)_{[n]}, \quad (5.4)$$

$$V_N(E)_{[n]} = \sum_{n_1+n_2+n_3=n} \beta_{2[n_1]} w_2(E)_{[n_2]} \beta_{2[n_3]}$$

is to be inserted into the model space Schrödinger equation (4.3). This provides the desired solution so that the perturbation theory based on the moment expansion (5.1) is completed.

The connection of our scheme based on Eq. (5.1) with the standard perturbation decomposition is the following: First, the straightforward insertion of $H = H_0 + \gamma V, \gamma \ll 1$, into $\langle m \rangle$ gives

$$\begin{aligned} \langle m \rangle_{[0]} &= \langle X_1 | H_0^m | X_1 \rangle, \\ \langle m \rangle_{[1]} &= \langle X_1 | H_0^{m-1} V | X_1 \rangle + \langle X_1 | H_0^{m-2} V H_0 | X_1 \rangle \\ &\quad + \dots + \langle X_1 | V H_0^{m-1} | X_1 \rangle, \\ &\dots \end{aligned} \quad (5.5)$$

$$\begin{aligned} \langle m \rangle_{[m]} &= \langle X_1 | V^m | X_1 \rangle, \\ \langle m \rangle_{[m+r]} &= 0, \quad r = 1, 2, \dots \end{aligned}$$

Second, the diagonality of H_0 (and $D_1 = 1$) implies,

$$\langle X_1 | H_0^k = \langle X_1 | H_0^k | X_1 \rangle \langle X_1 | \quad (5.6)$$

so that the knowledge of moments of the interaction (perturbation) V is a sufficient input for obtaining the (nonlinear) equation

$$\det[E_n - PHP - W_N(E_N)] = 0 \quad (5.7)$$

[i.e., Eq. (2.10)] for energies and Eq. (4.3) for the eigenvector projections. It is also possible (third) to work with the formal expansion $E = \sum_{n=0}^{\infty} \gamma^n E_{[n]}$, but this will not be done here. We may conclude that our formulas represent just the modification of the standard Brillouin-Wigner (BW) perturbation theory.¹

It is seen that the second assumption (diagonality of H_0 , i.e., specification of the basis $|X_k^i\rangle$ as eigenvectors of H_0) may be omitted in our approach for such H 's and H_0 's

that directly permit us to evaluate the matrix elements in Eq. (5.5), e.g., by analytical means when H_0 = kinetic energy = differential operator and V = local potential. In this way, we skip one step in the standard algorithm, namely the diagonalization of H_0 .

The MCF character of $V_N(E)$ resembles strongly the Padé approximation techniques¹⁷ (resummation¹⁸) and occurs here in the most natural way from the very beginning: We may place all the large matrix elements of H (e.g., kinetic operator T and any polynomial approximation to V in the oscillator basis) into H_0 and guarantee the smallness of the perturbation $\gamma V = H - H_0$, permitting H to be block-tridiagonal. Then, in the lowest order, the weakened form of Eq. (5.6), namely

$$PH_0^k = PH_0^k I_{k+1}, \quad (5.8)$$

completes the analogy with the modified BW theory because the formal extension of the model space projector $P \rightarrow I_{2N-1} = P_{\text{new}}$ and the knowledge of the first moments $I_m V I_n$, $m+n=2N$, of perturbation reduces the evaluation of $\langle m \rangle_{[0]}$ and $\langle m \rangle_{[1]}$ in Eq. (5.5) to the finite matrix multiplication again. In higher orders the exact matrix form of $\langle m \rangle_{[k]}$, $k \geq 2$, contains the infinite summations, but this is present in the BW case as well.

In the important first order case, the simplified form of Eq. (5.4) (the effective interaction correction) reads

$$\begin{aligned} V_N(E) = & \beta_{2[0]} w_2(E)_{[0]} \beta_{2[0]} + \gamma \beta_{2[1]} w_2(E)_{[0]} \beta_{2[0]} \\ & + \gamma \beta_{2[0]} w_2(E)_{[0]} \beta_{2[1]} + \gamma \beta_{2[0]} w_2(E)_{[1]} \beta_{2[0]}, \quad (5.9) \\ w_k(E)_{[1]} = & -w_k(E)_{[0]} \left[E \beta_{k[1]} - \alpha_{k[1]} \right. \\ & - \sum_{n_1+n_2+n_3=1} \beta_{k+1[n_1]} w_{k+1}(E)_{[n_2]} \\ & \left. \times \beta_{k+1[n_3]} \right] w_k(E)_{[0]}, \quad k=2,3,\dots,N. \end{aligned}$$

It should be complemented by the MCF [the first row of Eq. (5.3)] and by the definition of parameters [Eq. (5.2) for $n=0,1$ plus similar equation for α 's].

In practice, the cutoff N is to be fixed by the properties of H_0 but the classical dependence among H_0 , E_0 (be eigenvalue of H_0) and $|X_k\rangle$ (be eigenstates of H_0) is removed in the spirit of the realistic situation:

(a) The reliable (variational) initial approximation E_0 to E usually exists, shortening the root search in Eq. (5.7).

(b) The good physical models of $|\psi_N\rangle$ simplify the construction of model space and may lower significantly the cutoff N (cf. the truncation procedure mentioned at the end of Sec. 4).

(c) The rest of the unperturbed basis $|X_2\rangle, |X_3\rangle, \dots$ is almost arbitrary and the flexibility of partitioning enables us to keep the magnitude of perturbation $H - H_0$ within any predetermined limits. Thus, we may avoid the convergence problems and the necessity of using the higher perturbation orders for the standard two-particle interaction V even if it is not small as a whole. This is very useful since the higher moments V^k , $k \geq 2$, represent the more-particle forces.

6. A FEW REMARKS FROM THE NUMERICAL POINT OF VIEW

When we interpret H as the block-tridiagonal or finite matrix, then Sec. 2 or 4, respectively, defines the methods of its numerical diagonalization. Nevertheless, they are not designed to replace the standard computer codes for diagonalization of the matrix form of H . The reason is simple: From the computer time point of view they are less effective than their special case with $M=1$ suggested by Lanczos.⁶ The exceptions are as follows.

(a) We want to reduce the loss of precision ϵ . In the generalized Lanczos algorithm generating the basis $|X_k^i\rangle$ from $|X_1^i\rangle$ by H according to Eq. (2.6) (cf. also II), the value of ϵ is proportional to the number of subtractions²; it decreases as $1/M$ (at fixed dimension $n = M \times N$). In some cases, the necessity of reorthogonalizing² the basis may thus be avoided.

(b) H is a band matrix [e.g., from (a)]. Then, the parameters A_k, B_k , and C_{k+1} , $k=1,2,\dots$, are not free and are fully determined by the matrix elements of H^* (cf. I, especially Sec. 2). Here, Eq. (2.10) [the condition (D4)] has a standard meaning of the secular equation $\det(E - H) = 0$. It determines not only the eigenvalues λ_0 (cf. Ref. 19) but also MCF $F_k(\lambda_0)$ and hence the eigenvectors $|Y\rangle$ at the same time. Physically, the advantage lies in nonmixing of the different configurations in the basis which remains fixed.

(c) H is a sparse matrix and has the block-tridiagonal form after the basis is appropriately renumbered. This generalizes the band method (b). Some important technical details (regularization of random singularities, etc.) were discussed elsewhere.⁷

(d) H may be approximated by the block-tridiagonal matrix H_0 . E.g., the many-body Hamiltonian $H = T + V$ in the oscillator basis may be truncated in the nonstandard way preserving only the T matrix elements outside the finite (M -dimensional, $M < \infty$) model space. Then $H_0 = T + PVP$ is the block-tridiagonal matrix (the fanlike one, with $M_{k+1} > M_k$). MCF formulas represent a direct generalization of the ($M=1$) continued fraction method of Bassichis and Strayer.^{20,13} The perturbation theory of Sec. 5 contains an explicit prescription for including the corrections $V - PVP$ in a systematic way.

(e) The matrix H may be truncated to the block-tridiagonal form H_0 . The smallness of the error should be tested iteratively, increasing partition dimensions M_k (i.e., "opening the fan").

(f) The $n \times n$ matrix H is entirely general and it is too large to be stored in the computer. In this case, the choice of the model space may iteratively be improved in the spirit of the Lanczos method and in a way suggested for $M=1$ by Berger *et al.*²¹ They use the lowest cutoff $N=2$. The $M > 1$ generalization of this is quite straightforward: From the old basis $|X_1^k\rangle, k \in 1, M_1, |X_2^k\rangle, k \in 1, M_2$, we define the new vectors $|X_1^k\rangle_{\text{new}} = |Y\rangle = \mathcal{N}|\psi_0\rangle$ according to Eq. (2.9), and arbitrary $|X_2^k\rangle_{\text{new}}$ orthogonal to $|X_1^k\rangle_{\text{new}}$. All the necessary nonzero parameters A_1^j, B_1^j, C_2^j , and A_2^j are defined by the lowest moment formulas (II) adapted for convenience as

$$\begin{aligned} PHP &= A_1 (= \alpha_1), \\ PHQHP &= B_1 C_2 (= \beta_2), \quad Q = 1 - P, \\ PHQHQP &= B_1 A_2 C_2 (= \alpha_2). \end{aligned} \quad (6.1)$$

As well as for $M = 1$, this definition of A_1 , B_1 , C_2 , and A_2 is not unique and admits some convenient conventions (cf. $C_2 = B_1^+$ and the choice of positive sign in Ref. 21).

(g) In paragraph (f), entering of $QHQP$ $\{[(n - M) \times (n - M)]$ -dimensional matrix $\}$ into the storage of the computer and the matrix multiplication on the left-hand side of Eq. (6.1) are to be done "row-wise" since we assume $n^2 \gg$ storage capacity $> n \gg N \times M$. Hence, the restriction to $N = 2$ is not substantial. Using the general moment formulas (II) in the form of the algorithm initialized by the three $M \times M$ matrices $g_1 = 0$, $\beta_1 = 1$, $\alpha_1 = PHP$, two $(n - M) \times M$ matrices $T_0 = 0$, $W_1 = QHP$, and two $(n - M) \times (n - M)$ matrices $Y_0 = 1$, $U = QHQ$ (to be manipulated row-wise), we define the auxiliary sequences α_k and β_k by the recurrent prescription

$$\begin{aligned} g_{k-1} &= (\beta_{k-1})^{-1}, \\ Y_{k-1} &= Y_{k-2} - T_{k-2} g_{k-1} T_{k-2}^+, \\ T_{k-1} &= Y_{k-1} W_{k-1}, \quad \beta_k = W_{k-1}^+ T_{k-1}, \\ W_k &= U T_{k-1}, \quad \alpha_k = T_{k-1}^+ W_k, \quad k = 2, 3, \dots, N. \end{aligned} \quad (6.2)$$

The first three items may be skipped for $k = 2$ when we change the initialization to $g_1 = 1$, $T_1 = W_1$, $Y_1 = 1$. It may be shown that the algorithm (6.2) is an equivalent reformulation of the method of Sec. 4 when the moments of H are not known and are to be approximately evaluated by the (truncated) matrix multiplication.

7. CONCLUSIONS

The main feature of our method is its structure of the Born series reordered by means of the introduction of free parameters (idea of Haydock²²). In the present third part of the series (I, II), we may conclude that it treats both the homogeneous and nonhomogeneous equations in the same way.

The flexibility of the method is connected with the *a priori* freedom in the choice of parameters. In this sense, we may interpret and relate the special cases (Taylor series + orthogonalization = Lanczos method⁶) as well as the

generalizations ($M = 1$ Lanczos method $\rightarrow M > 1$ Graffi and Grecchi¹⁹ band matrix MCF method). It seems to us that the practical value of such a general formalism is in filling the gaps between the known methods (e.g., between the classical theory of moments¹² and the matrix of Padé approximants²³ to the Born series or the generalized Brillouin-Wigner perturbation theory²⁴), but the main merit will be its heuristic power. We believe that it is well illustrated by our generalizations of the Bessel functions ($M = 1$ power series $\rightarrow M > 1$ Dirichlet series²⁵), by the model space formulas for the effective interaction corrections, and by a new formulation of the perturbation theory. Because of the methodical character of the paper, we have deferred further applications as well as any detailed calculations to later publications.

¹K. Kumar, *Perturbation Theory and the Nuclear Many Body Problem* (North-Holland, Amsterdam, 1962).

²J.H. Wilkinson, *The Algebraic Eigenvalue Problem* (Clarendon, Oxford, 1965).

³M. Znojil, J. Math. Phys. **18**, 717 (1977).

⁴M. Znojil, J. Math. Phys. **20**, 2330 (1980).

⁵R.G. Newton, *Scattering Theory of Waves and Particles* (McGraw-Hill, New York, 1966).

⁶C. Lanczos, J. Res. NBS **45**, 255 (1950).

⁷M. Znojil, J. Phys. A: Math. Gen. **11**, 1501 (1978).

⁸E. Kamke, *Differentialgleichungen I* (Leipzig, 1959).

⁹A. Martin, Nuovo Cimento **14**, 403 (1959); L. Trlifaj, J. Math. Phys. **21**, 1724 (1980).

¹⁰I.S. Gradshteyn and I.M. Ryzhik, *Tablitsy Integralov, Summ i Proizv* (Nauka, Moscow, 1971).

¹¹P.O. Löwdin, J. Math. Phys. **3**, 969 (1962).

¹²N.I. Akhiezer, *The Classical Moment Problem* (Oliver and Boyd, Edinburgh, 1976).

¹³M. Znojil, J. Phys. A: Math. Gen. **9**, 1 (1976).

¹⁴H. Feshbach, A. Kerman, and S. Koonin, "Theory of statistical multistep reactions," preprint 1977, to be published.

¹⁵R.R. Whitehead and A. Watt, J. Phys. G: Nucl. Phys. **4**, 835 (1978).

¹⁶E.g., R.R. Whitehead, Nucl. Phys. A **182**, 290 (1972).

¹⁷G. Baker, *Padé Approximants in Theoretical Physics* (Academic, New York, 1970).

¹⁸J. Killingbeck, Rep. Prog. Phys. **40**, 963 (1977).

¹⁹S. Graffi and V. Grecchi, Lett. Nuovo Cimento **12**, 425 (1975).

²⁰W.H. Bassichis, M.R. Strayer, Phys. Rev. C **18**, 1505 (1978).

²¹W.A. Berger, H.G. Müller, K.G. Kreuzer, and R.M. Dreizler, J. Phys. A: Math. Gen. **10**, 1089 (1977).

²²R. Haydock, J. Phys. A: Math. Nucl. Gen. **7**, 2120 (1974).

²³G. Turchetti, Fortschr. Phys. **26**, 1 (1978).

²⁴G. Turchetti, F. Ortolani, and C. Sagretti, Nuovo Cimento A **44**, 211 (1978).

²⁵A.F. Leontiev, *Ryady Exponent* (Nauka, Moscow, 1976).